LUMBINI ENGINEERING COLLEGE

(Affiliated to Pokhara University)

Final Internal Exam

2082

Level: Bachelor
Programme: BE (Computer)
Course: Artificial Intelligence
Semester: V (Fifth)

Year : 2082
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

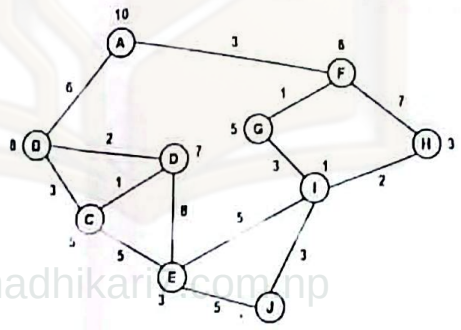
1. a) Justify that "System that think rationally" and "System that act rationally" are the part of artificial intelligence. Explain it with practical examples. [7]

   b) In problem solving, why problem formulation must follow goal formulation? Using constraint satisfaction problem, solve the following crypto-arithmetic problem. [8]

   CRYPTO + LOGIC = SECURE

2. a) Design a goal-based agent for an automated taxi. Describe its sensors, actuators, and how it uses goals to make decisions. [8]

   b) What is A* Search? Find the most cost-effective path to reach from start state A to final state J using A* search Algorithm. [7]



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3. a) The logical operator "←→is read "if and only if". P ←→Q is defined as being equivalent to (P → Q) ∧ (Q→P). Based on this definition, show that P ←→ Q is logically equivalent to (P ∨ Q) →(P ∧ Q) using truth table. [8]

   b) Describe semantic networks and their role in knowledge representation. Create a semantic network for a specific domain, such as animals and their characteristics. [7]

4. a) Discuss the potential impact of machine learning on various industries, such as healthcare, finance, and autonomous vehicles. [8]

   b) Represent the following statements in First-Order Predicate Logic [7]

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a. Every child loves every kitten.
b. No manager likes an employee who is lazy.
c. There is a politician who represents all citizens.
d. Only one person can be the president of a club.

5. a) What is fuzzification? Explain in detail all the fuzzification methods with suitable examples. [7]

b) Cluster the following Seven points into THREE clusters (K=3) using the K-Means algorithm. Use A1(3, 11), A2(3, 6), A3(9, 5), A4(6, 9), A5(6, 4), A6(5, 3), A7(2, 3)) as initial cluster centers. Show only the first iteration of assigning points to clusters and calculating the new centroids. [8]

6. a) Explain how a single layer perceptron learn logical **OR** operation. Assume weight W_1=0.24 and W_2=0.49 and learning rate **alpha** = 0.14 [8]

b) Explain the architecture of a typical Expert System. Describe the functions of its main components: Knowledge Base, Inference Engine, User Interface. [7]

7. Write short notes on: (Any two) 2×5

a) Machine Learning
b) Inference and Reasoning
c) Total Turing Test

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POKHARA UNIVERSITY

School of Engineering

INTERNAL ASSESSMENT

Level: Bachelor | Semester: Spring | Year : 2026

Programme: BE | Full Marks: 100 | Pass Marks: 45

Course: Artificial Intelligence | Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is artificial intelligence? Explain the components of intelligence in detail. 8
   b) What is an intelligent agent? Explain the simplex intelligent agent in detail. 7
2. a) How does the A* search resolve the problem of **greedy best first search** that may stuck in a loop? Explain with example. 7
   b) What is an adversarial search? Explain the min-max algorithm with an example. 8
3. a) Explain the constrain satisfaction problem with the example of map coloring problem. 8

OR

Explain backtracking search in the constrain satisfaction problems.

   b) What are the drawbacks of propositional logic? How are they resolved in predicate logic? Explain in detail. 7
4. a) Define reasoning under uncertainty. Explain a Bayesian Network with an suitable example. 5
   b) Consider the following facts: 10

1. All hounds howl at night.
2. Anyone who has any cats will not have any mice.
3. Light sleepers do not have anything which howls at night.
4. John has either a cat or a hound.
5. John is a light sleeper.

Using resolution prove that John does not have any mice.

5. a) What is linear regression? Explain with a suitable example. 8
   b) Define Artificial Neural Network. Explain how an ANN work in detail. 7
6. a) Define a fuzzy logic. Explain the components of fuzzy inference system. 8

OR

Explain a Mamdani fuzzy inference system with any suitable example.

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b) Explain the importance and need of of an expert system. 7

7 Write short notes on any two (2X5) 10

a) Acting Humanly

b) Simulated Annealing

c) Semantic Nets

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GANDAKI COLLEGE OF ENGINEERING AND SCIENCE

Level: Bachelor - Semester: Fall Year : 2025
Programme: BE CE V Full Marks: 100
Course: **Artificial Intelligence** Pass Marks: 45
Time : 3hrs.

The figures in the margin indicate full marks.

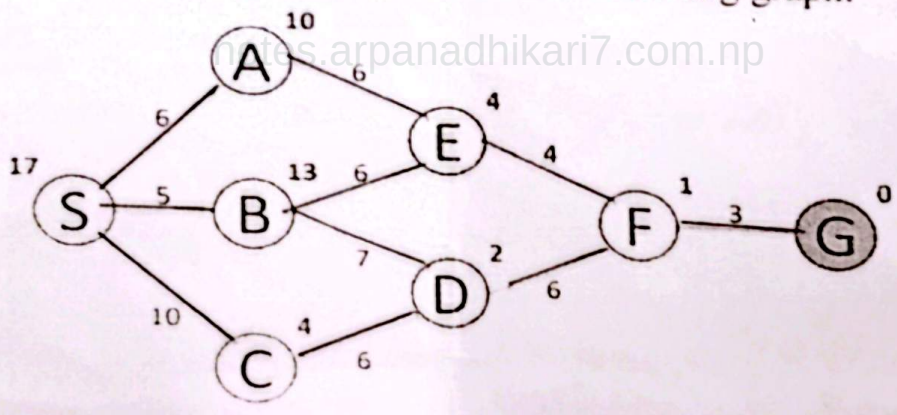
Attempt all the questions.

1. a) Justify that "system thinks rationally" and "System acting humanly" are the part of artificial intelligence. Explain with examples. 8

   b) What is agent function? Explain simplex reflex and model based agents with examples. 7

2. a) In a distinct land, bigamy is common. There are six people who want to cross a river in this land. This group consists of two men, each two wives. No man can tolerate any of his wives being in the company of another man unless yet least he or his next wife is present in the boat or in next land. There is a boat that holds two people to be used for crossing the river. How is the trip possible? Formulate the problem. 8

   b) Perform A* and greedy search on the following graph. 7



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3. a) Convert $B \leftrightarrow (P \vee Q)$ into conjunctive normal form(CNF). 7

   b) Represent following sentences with predicate logic. 8
      i. Every gardener likes the sun.
      ii. All purple mushrooms are poisonous.

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iii. No purple mushrooms are poisonous.
iv. A car have at least one door.

4. a) How does k nearest neighbor algorithm works? Explain it with suitable example. 8

b) Write down the k- means clustering algorithm. Cluster the given 8 points in 3 clusters 7
A(2,10), B(2,5),C(8,4),D(5,8),E(7,5),F(6,4),G(1,2),H(4,9).

5. a) Explain fuzzification and defuzzification with suitable examples. 8

b) Explain perceptron learning theory to train AND logic. 7

6. a) What is expert system? Explain architecture of expert system with necessary diagram. 8

b) Represent the following sentences into a semantic network. 7

i. Birds are animals.
ii. Birds have feathers, fly and lay eggs.
iii. Albatross is a bird.
iv. Donald is a bird.
v. Tracy is an albatross.

7 Write short notes on **any two**: 2x5

a) Constraint satisfaction (crypto-arithmetic) problem

b) Baysian Network

c) Forward and backward chaining

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Pokhara University

Everest Engineering College

Final Internal Assessment
Fall-2082

Level: Bachelor — **F.M.** 100
Program: BE CMP (5th Sem) — **P.M.** 45
Subject: Artificial Intelligence — **Time:** 3hrs

Attempt all the questions.

1 a) Difference between Turing test and Total Turing test. Justify that "System that think rationally" and "System that act rationally" are the part of artificial intelligence. Explain with example. 2+6

b) Define agents and environment? How is the design of an intelligent agent affected by the type of environment (deterministic vs. stochastic, fully observable vs. partially observable)? 7

2 a) Define a Constraint Satisfaction Problem (CSP). Solve the following crypto-arithmetic problem using CSP techniques, where each letter represents a unique digit (0-9), and identical letters represent the same digit: 8

SEVEN+EIGHT = TWELVE

b) What are the evaluation strategies of the A* algorithm? Solve the 8-puzzle problem using the A* algorithm. 7

Start

2	8	3
1	6	4
7		5

Goal

1	2	3
8		4
7	6	5

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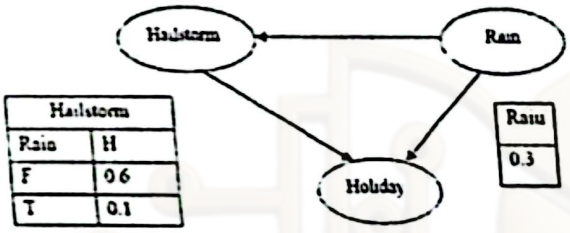
3 a) Explain the concept of alpha and beta value in the alpha beta pruning algorithm. How does these values propagate through the search tree? Show with an example. 7

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b) Assume the following facts: 8

Horses, cows, pigs are mammals.
An offspring of a horse is a horse.
Bluebeard is a horse.
Bluebeard is a Charlie's parent.
Offspring and parent are inverse relations.
Every mammal has a parent.
Prove that "Charlie is a horse" using resolution.

4 a) Differentiate between semantic nets and frames. Analyze the given graph and find the probability of holiday given that there is rain, and hailstorm. 3+4



Hailstorm	
Rain	H
F	0.6
T	0.1

Rain
0.3

Holiday		
Hailstorm	Rain	H
F	F	0.0
F	T	0.7
T	F	0.8
T	T	0.99

b) Explain the steps involved in Conjunctive Normal Form (CNF). Discuss the inference rule for propositional logic with example. 8

5 a) What are the labelled and unlabeled data? Suppose you are provided with unlabeled data to train a machine learning model. Which machine learning model will you apply to train the learning model? Explain in detail with example. 7

b) Write the Backpropagation algorithm. Demonstrate its use in designing an XOR gate through one iteration. 8

6 a) Define membership functions. Discuss the method of fuzzification and defuzzification with an example. 7

b) What is an expert system? Explain the architecture of an expert system. 8

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Universal Engineering & Science College

Affiliated to Pokhara University

Chakupat, Lalitpur

Level: Bachelor Semester: 5th Year : 2026

Programme : BE Computer Time : 3 hours Full Marks: 100

Subject:Artificial Intelligence Set: A Pass Marks: 45

Pre-Board Examination -2082 (Fall 2025)

Candidates are required to give their answer in their own words as far as possible.
The figure in the margin indicate full marks.

Attempt all questions

1 a)Define intelligence.Explain turing test . (7)

b)Define state space problem in solving.Using constraint satisfaction problem,solve the following crypto arithmetic problem. (8)

BASE+BALL = GAMES

2 a)Define intelligent agent .define goal based and utility based agent with suitable diagram and example. (7)

OR

What is rational agent?Give any two examples of PEAS explaining what is PEAS.

2b)Explain depth first search and breadth first search with its completeness,optimality,time complexity and space complexity (8)

3a) Write difference between A* search algorithm and Best First search algorithm. (7)

3b)Describe the following features of genetic algoriths(GA) encoding,fitness function,selection,crossover and mutation.Explain with any sutable example. (8)

4a) Given the propositional statements p: "It is raining" and q: "The road *is wet", write the logical expressions* for the following. [7]

i. If it is not sunny, then the picnic will not happen.

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ii. If it is sunny, then the picnic will happen, and if the picnic happens, then it is sunny.

iii. It is not the case that both it is sunny and the picnic will happen.

iv. If the picnic happens, then it is sunny, but if it is not sunny, then the picnic will not happen

4b)Using the truth table prove that ¬(p∧q) is equivalent to ¬p∨¬q [8]

5a)Assume the following facts [7]

All birds have wings.

Penguins are birds.

No animals with wings are mammals.

Tweety is a penguin.

Use resolution to prove that Tweety is not a mammal.

5b)What is rote learning? Differentiate between Supervised and Unsupervised Learning. [8]

OR

What is a multilayer Neural Network? Implement simple logic XOR using Backpropagation in Neural Network.

6a) Explain the advantages and limitations of an expert system. Explain the architecture of the expert System. [7]

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6b) Describe briefly about Logistic regression and k Means Clustering in Machine Learning. [8]

7. Write Short notes any two [2*5]

i) simulated annealing

ii)Natural Language Processing

iii) Semantic Nets

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UNITED TECHNICAL COLLEGE

Level: Bachelor
Programme: BE
Course: Artificial Intelligence
Semester: 5th

Semester: Fall

Year : 2025
Full Marks : 50
Pass Marks : 25
Time :1.5hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt any three questions from 1-6. Question No.7 is Compulsory

1 a. Define Artificial Intelligence (AI). Explain its application with suitable examples. 7

b. Explain the concept of rationality in intelligent agents. What are task environments and their properties? 8

2 a. Discuss and compare the uninformed search techniques: BFS, DFS, and Iterative Deepening Search. 7

b. Explain the A* search algorithm with an example. 8

3 a. Explain the working of genetic algorithms. 7

b. Explain the Minimax algorithm with an example of game-playing. 8

4 a. Assume the following facts: 8

i. John likes all kinds of food.
ii. Apple and vegetable are food
iii. Anything anyone eats and is not killed is food.
iv. Anil eats peanuts and still alive
v. Harry eats everything that Anil eats.
vi. Anyone who is not killed means, he is alive.

Represent these facts in predicate logic. Using Resolution prove "John likes Peanuts"

b. Define Semantic Network. Draw semantic network of following clauses: 7

Subset_of(Human, Mammal), Subset_of(Male, Human), Subset_of(Female, Human), Has_Mother(Human, Female), Member_of(Mary, Female), Member_of(John, Male), Husband_of(John, Mary)

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5 a. Define machine learning. Explain the K-means clustering with an example. 7

b. Explain backpropagation in neural networks. Why is the activation function important? 8

6 a. Explain the steps of fuzzification and defuzzification with examples. 7

b. Define expert systems. Explain their architecture and components with a diagram. 8

7 Write short notes on(Attempt Any Two): 5

a. Impact of AI on employment, privacy, and security

b. Linear Regression

c. CSP

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NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Level: Bachelor Assessment
Programme: BE (Computer Day/Morning)
Course: Artificial Intelligence (New)

Year : 2025
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

✔ *Candidates are requested to give their answer as far as practicable in their own words.*
✔ *The figure in the margin indicates the full marks*
✔ ***Attempt ALL question***

1) a) What is an intelligent system? Also, what is acting humanly, Explain Turing Test with an example. [7]

b) What is a rational agent? Compare between model based and goal based agent with an example. Also find the PEAS of self-driving car [8]

2. a) Differentiate between uninformed and informed search. What are the limitation of A* search. Explain with examples. [7]

b) What is a genetic algorithm? Maximize the function $f(x) = x^3$ using a genetic algorithm. [0,31] [8]

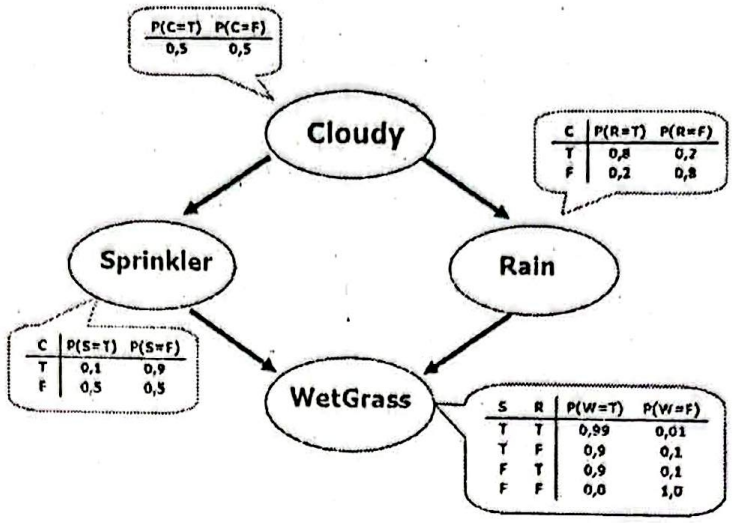
3. a) What is gradient descent? Explain how hill climbing algorithm works. [7]

b) What is predicate logic? For the given set of knowledge, convert them into FOPL. [8]

- Every Apple is either green or red.
- No apple is blue
- Some people like garlic.
- All basketball players are tall.

4. a) What are Bayesian Networks? For the given graph, calculate the probability that there is wet grass even if there was no cloud, no sprinkler on and no rain. [7]

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b) Explain supervised learning with examples. For the following data using K=5, determine the customer type for purchase amount 60.5 and Items in cart 3. [8]

Purchase Amount ($)	35.5	80.25	15.99	120.5	42	65.75	22.9
Items in Cart	2	4	1	6	3	3	1
Customer Type	New	Returning	New	Returning	New	Returning	New

5. a) What is a perceptron? Compare feed forward and feed backward networks with examples. [7]

b) What is learning rate? Create a perceptron network for OR Logic Gate. Consider learning rate, activation function on your own. [8]

6. a) What is fuzzy logic? Explain in detail the Mamdani method for fuzzy inference systems? [7]

b) What is an expert system? Explain its architecture and describe forward, backward chaining methods with examples. [8]

7. Write a short note: (Any two) [10]

a) Frame b) Hidden Markov Model c) Alpha Beta Pruning

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National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

First Term Examinations

Level: Bachelor | Semester: Fall_V | Year : 2025

Program: B.E. Computer | F.M. : 100

Course: **Artificial Intelligence** | P.M. : 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

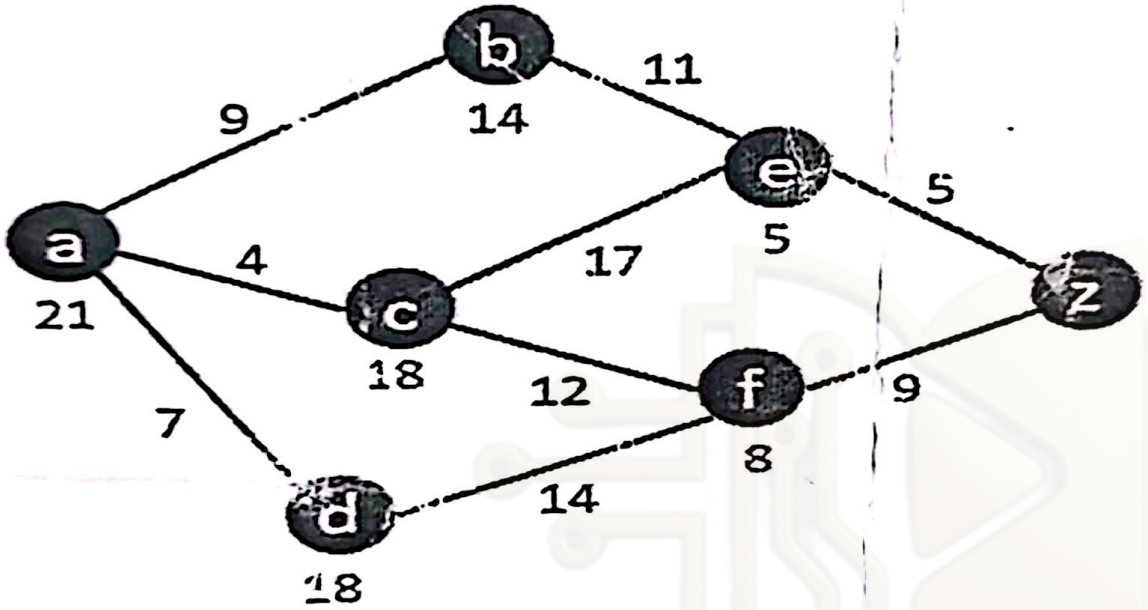
Attempt all the questions.

1	a.	What are different types of Intelligence? What is the importance of AI?	8
	b.	AI is replacing a lot of automation tasks that used to be done by humans, do you think AI will replace humans entirely?	7
2	a.	What are different types of Agents. Explain Model based Agent with diagram.	8
	b.	Describe the foundations of AI. How do fields like psychology, philosophy, and computer science contribute to AI development?	7
3	a.	What is the relevance of cognitive modeling in the "Thinking Humanly" approach? Explain with examples	8
	b.	What types of problem are categorized as well-defined problems.	7
4	a.	What do you mean by Turing Test? Explain in detail.	7
	b.	Describe the PEAS (Performance measure, Environment, Actuators, Sensors) representation of a task environment. Provide any real-world examples.	8
5	a.	How do learning agents improve over time? Discuss with a real-world AI system.	7
	b.	What is simulated annealing? Explain how it avoids getting trapped in local maxima?	8

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6 a. Solve the following using A* approach. Take A as starting node. 7



b. Write algorithm for both DFS and BFS. What types of data structure is used by them 8

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7 Write shorts notes on (any two): 10

a. Hill climbing Algorithm
b. Informed and uninformed search
c. Game Playing

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National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

Pr-University Examinations

Level: Bachelor | Semester: V Fall | Year : 2025

Program: BE Computer | F.M. : 100

Course: **Artificial Intelligence** | P.M. : 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1 a) Can AI be conscious? Like can it have something like pain, awareness, emotions, selfhood? Justify. 7

b) Explain the four approaches to AI: Acting Humanly, Thinking Humanly, Thinking Rationally, Acting Rationally. 8

2 a. Describe PEAS representation with a real-world example. 8

b. Differentiate well-defined vs ill-defined problems with examples 7

3 a. Define CSP. Explain variables, domains, constraints, and backtracking search. Solve ABC + DEF = GHI. 9

b. Compare IDS with DFS and BFS. 6

4 a. Solve a search problem using A* algorithm with heuristic values: h(A)=10, h(B)=8, h(C)=5, h(D)=7, h(E)=3, h(F)=0 and Edges: A→B(6), A→C(3), B→D(2), B→E(4), C→E(7), C→F(10), D→F(1), E→F(5). With start node as A and goal node is F. 7

b. Convert the following statements into Propositional Logic and CNF then Prove the conclusion using Resolution. 8

i. If the system is secure, then data is protected.

ii. If data is protected and backups exist, then recovery is possible.

iii. If recovery is possible, the company continues operation.

iv. The system is secure.

v. Backups exist.

Prove: The company continues operation.

5 a. Perform K-means clustering (k=2) Points: (2,10), (2,5), (8,4), 8

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(5,8), (7,5), (6,4). Take initial centroids as: (2,10), (5,8). What are applications of K means algorithm?

b. Explain Artificial Neural Network architecture and biological inspiration. 7

6 a. Explain Architecture of Expert System. 7

b. For given Fuzzy sets, A = {(1,0.2), (2,0.5), (3,0.9)} and B = {(1,0.7), (2,0.4), (3,0.3)}. Find: Union, Intersection, Complement 8

7 Write short notes on (any two) 1(

a. Turing test VS Rational Agent

b. Rule based Systems

c. Backpropagation

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24 February 2026 2:56 pm

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POKHARA ENGINEERING COLLEGE
INTERNAL ASSESSMENT

Level: Bachelor | Semester – Fall | Year : 2025
Programme: B.E. | Full Marks: 100
Course: Artificial Intelligence | Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1 a What is ethics in AI? Explain about the ethical implications of AI? 8

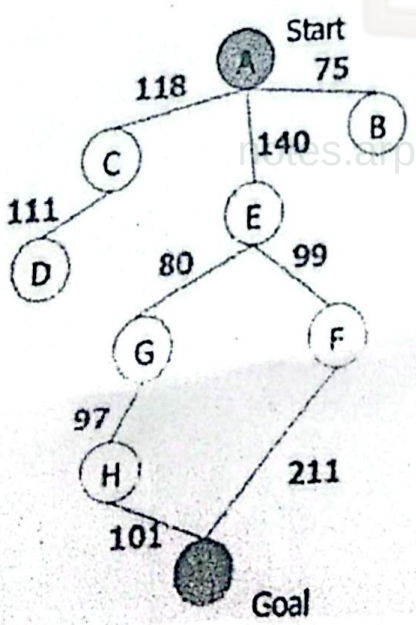
b Define agent function. Differentiate between Utility based agent and Goal based agent with necessary examples. 7

2 a Solve the following cryptographic problem: 7

$$\begin{array}{r} W\ R\ O\ N\ G \\ +\ W\ R\ O\ N\ G \\ \hline R\ I\ G\ H\ T \end{array}$$

b What is informed and uninformed search? How DLS overcomes the limitations of DFS, explain with appropriate diagram. 8

3 a Calculate the path cost using greedy first search and A* search. 8



State	Heuristic: h(n)
A	366
B	374
C	329
D	244
E	253
F	178
G	193
H	98
I	0

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b Why do we need backpropagation algorithm? Explain its working principle with necessary diagrams. 7

4 a Define Genetic Algorithm. Explain the operators in Genetic Algorithms. 8

OR

Explain the significance of pruning in game playing algorithm. How does alpha-beta pruning work and how does it improve the efficiency of the search process in minimax algorithm?

b Given the following statistics, what is the probability that a woman has cancer id she has a positive mammogram result? 7

i. One percent of women over 50 have breast cancer.

ii. Ninety percent of women who have breast cancer test positive on mammograms.

iii. Eight percent of women will have false positives.

5 a What is Expert system. Explain case Based Reasoning with necessary diagram. 8

b Tom is cat. Tom caught bird. Tom is owned by John. Tom is ginger in color. Cats like cream. The cat sat on the mat. A cat is a mammal. A bird is an animal. All mammals are animals. Mammal have fur. Draw a semantic graph. 7

6 a Represent the following statements into predicate logic: 8

i. All big cars are expensive

ii. There are some small cars that are expensive

iii. A car is expensive only if it is big.

iv. All cars have at least one door.

b What is fuzzy logic? Explain the architecture of fuzzy logic system. 7

7 Write short notes on: (Any Two): 5x2 =10

a. Alpha Beta Pruning

b. Support Vector Machine

c. Fuzzy Inference System

The End

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NEPAL ENGINEERING COLLEGE

Level: Bachelor 2025 – Fall Batch : 2023
Programme: BE (Computer) Full Marks: 100
Pass Marks: 45
Course: Artificial Intelligence Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What is Artificial Intelligence? Discuss the impact of artificial intelligence in modern society. 2+5
   b) Differentiate intelligent agents and rational agents. What is PEAS? Discuss PEAS for *Vacuum Cleaning Agent*. 3+5
2. a) What are the components of a state space search problem formulation? Formulate a search problem for 8-Puzzle, discuss production rules, and provide a solution. 3+4
   b) What is Cryptarithmetic Problem? Formulate it as CSP and provide a solution for the following cryptarithmetic problem 3+5

   SUN + FUN = SWIM
3. a) State and discuss Hill Climbing Algorithm. What are the issues associated with the Hill Climbing Problem, demonstrate with the example of block arrangement problem. 3+4
   b) State Simulated Annealing Algorithm? The following table shows three evaluations of a simulated annealing algorithm. For each evaluation calculate the probability of the next state being accepted. Assume the objective function is being maximized. 4+4

Current Evaluation	Neighbourhood Evaluation	Current Temperature
80	65	10
80	65	100
80	65	500

   Ensure you show the formula you use and describe the terms.
4. a) Prove the following using Resolution Refutation System: 8
   All those who honour both their parents are blessed. If anyone dislikes any of his siblings, one does not honour one's parents. Raju likes his sister Rita. Therefore, Rnju is blessed.
   b) What is a partition semantic net? Represent the following sentences in a semantic network: *Every dog had bitten a postman* 3+4

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5. a) What are the steps to convert a First-Order Predicate Logic (FOPL) sentence to clausal form? Provide a detailed step-by-step procedure, including explanations of key transformations for the following: 2+5

Consider the sentence: "Everyone who loves all animals is loved by someone."

In FOPL: $\forall x\,[\forall y\,(\text{Animal}(y) \Rightarrow \text{Loves}(x,y)) \Rightarrow \exists z\,\text{Loves}(z,x)]$ 2+6

b) What is Machine Learning? Discuss the difference between supervised learning, and unsupervised learning (*At least 6 points*).

6. a) Discuss the mathematical model of Neuron. State the perceptron training algorithm and demonstrate the same for AND Gate. 3+5

b) State and discuss the linear regression algorithm. Find the value of Y when X = 7 7

X	Y
1.0	1.3
1.5	1.6
5.0	4.8
8.0	7.6
2.0	2.5
9.0	8.2

OR

c) What is fuzzy logic? Explain its core concepts and discuss the main fuzzy set operations (union, intersection, and complement) with examples.

7. Write short notes on (Any Two): 2×5

a) Discuss MiniMax with example

b) A* Algorithm

c) Frame

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2+5

Nepal Engineering College

Level: Bachelor

Year:2026

Full Marks: 100

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COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test II

Date: 2082/11/11			
Level	BE	Full Marks	50
Programme	BCE	Time	
Semester	V	1.5 hrs	

***Subject:** Artificial Intelligence*

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ ***Attempt All question.***
- ✓ *The figures in the margin indicate **Full Marks**.*
- ✓ *Assume suitable data if necessary.*

1 a. Create a semantic Net and Frame system with following description:
"Ruby is a dog. Tommy is a tiger. Coco is a parrot. Parrot is a bird. Dog and Tiger are mammals. Both birds and mammals are animals. Birds have feathers and can fly. Mammals have fur and walk. Some animals live on land, and some live in water. Ruby and Coco are best friends." [8]

b. What are the sources of uncertainty? how does probabilistic reasoning, specifically Bayesian networks, help in reasoning under uncertainty? Explain with an example. [7]

2 a. What is machine learning? How supervised learning differs from unsupervised learning? [8]

b. Define Artificial Neural Network. Discuss about the back-propagation algorithm with example. [7]

3. a. Explain the concept of fuzzy sets and the role of membership function in fuzzy logic. [8]

b. What is the role of Knowledge Engineer in knowledge acquisition for an expert system? [7]

OR

Explain the architecture of expert systems in detail.

4. Write short notes on: **(Any two)** [2.5 x 2 =5]
   a. Predicate Logic
   b. K-means clustering
   c. Hidden Markov Model

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COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test I

Date: 2081/09/25			
Level	BE	Full Marks	50
Programme	BCE	Time	
Semester	V	1.5 hrs	

Subject: - Artificial Intelligence

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *Attempt All questions.*
- ✓ *The figures in the margin indicate Full Marks.*
- ✓ *Assume suitable data if necessary.*

1 a. Define Intelligence. Explain the current trends of AI and its impact on the society. [7]
b. Explain Turing Test. Describe the task environment of an autonomous vehicle. [8]

2 a. What is heuristics? Explain Alpha-Béta Pruning with an example. [7]
b. Explain Hill Climbing Algorithm. What are the problems in hill climbing algorithm? Also give the solutions for each problems. [8]

3 a. Assume the following facts.

1. If you study hard and attend all lectures, then you will pass the exam.
2. You will study hard if you solve all assignments or get help from a tutor.
3. You attended all lectures for this course.
4. You solved all the assignments given by your instructor.
5. You did not get any help from a tutor.

Based on these facts, prove that you will pass the exam [5]

4. a. Devise the truth table for $(p \rightarrow q) \rightarrow (\neg q \rightarrow \neg p)$ [5]
b. Write down the mathematical expressions using predicate logic: [10]

- Tom is a cat.
- Tom caught a bird.
- John owns Tom.
- Tom is ginger in color.
- Cats like cream.
- The cat sat on the mat.
- A cat is a mammal.
- A bird is an animal.
- All mammals are animals.
- Mammals have fur.

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Nepal College of Information Technology
Unit Test 2025-Fall

Program: BECE
Semester: III
Subject:Artificial Intelligence

Time: 2 Hours
FM: 70
PM: 35

- ✓ *Candidates are requested to give their answer as far as practicable in their own words.*
- ✓ *The figure in the margin indicates the full marks*
- ✓ *Attempt ALL question*

All questions carry equal marks(7

1. Define artificial intelligence and explain its four major approaches. How does each approach attempt to simulate or replicate human intelligence?
2. What are the major foundations of AI? Discuss how fields like philosophy, psychology, and computer science contribute to AI
3. Explain the benefits and potential risks associated with AI. How should society and governments respond to these risks?
4. What is a rational agent? Explain how performance measures, environment, and design influence rationality.
5. Describe the different types of agent programs. Which type would be most suitable for an AI-based recommendation system, and why?
6. Write the production rules to solve the classic Water Jug problem with a 3-liter and a 4-liter jug. How can we get exactly 2 liters of water in the 4-liter jug?
7. Compare uninformed and informed search strategies. Also, write the A* algorithm and explain its advantage over uninformed search methods.
8. Solve the following cryptarithmetic problem using constraint satisfaction techniques: SEND + MORE = MONEY
9. Using the Genetic Algorithm, maximize the function $f(x) = 15x - x^2$ in the range [0, 15]. Explain the steps involved in the process.
10. Explain how the minimax algorithm and alpha-beta pruning work in adversarial search. How does alpha-beta pruning improve the performance of game-playing agents?

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Madan Bhandari College of Engineering
Urlabari-3, Morang
Final Internal Examination

Level: Bachelor
Programme: B.E. (Computer)
Year/Part: III/I

Full Marks: 100
Pass Marks: 45
Time: 3 hrs.

Subject: - Artificial Intelligence (SET A)

- ✓ *Candidates are required to give their answers in their own words as far as possible.*
- ✓ *Attempt all questions*

1. a). What is Artificial Intelligence? Explain the current trends of AI and its impact on Society.
   b). Define agent function? Differentiate goal-based and utility-based agents with examples.
2. a). What are the constraints Satisfaction Problems? Solve the following Crypt-arithmetic Problem. 7

	B	A	S	E
+	B	A	L	L
G	A	M	E	S

   b). Explain with an Example how A* search can find the optimal solution when there is more than one solution? 8
3. a). What are the problems of depth first search and breadth first search? How are these problems resolved? Explain with a suitable example? 7
   b). Represent the following facts in predicate logic: 8
   i) John likes all kind of food.
   ii). Apple and vegetable are food.
   iii). Anything anyone eats and not killed is food.
   iv). Anil eats peanuts and still alive.
   v). Harry eats everything that Anil eats.
4. a). Consider the facts given in Q3 b) using resolution prove that John likes peanuts. 7
   b). What is machine learning? Discuss the difference between regression, classification and clustering in machine learning. 8
5. a). Explain the k-means clustering algorithm with step-by-step demonstration on a small dataset, showing how the result. 7
   b). How does support vector machine work? Illustrate with a suitable example? 8
6. a). Define the role of bias in a perceptron. How are weights updated in a single layer perceptron? Explain with example. 7
   b). what is Expert System? Explain the architecture of expert system in detail 8
7. Write short notes on [Any Two] 10
   a. fuzzy logic
   b. Semantic nets
   c. Learning by analogy

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Madan Bhandari College of Engineering
Urlabari-3, Morang
Final Internal Examination

Level: Bachelor
Programme: B.E. (Computer)
Year/Part: III/I

Full Marks: 100
Pass Marks: 45
Time: 3 hrs.

Subject: - Artificial Intelligence (SET B)

- ✓ *Candidates are required to give their answers in their own words as far as possible.*
- ✓ *Attempt all questions*

1. a). What is Artificial Intelligence? Have you ever used any AI technology like chatGPT? Do you think that it has intelligence? If yes, explain in detail.
   b). Explain the terms agent, percept, percept sequence, agent function and agent program with example of vacuum-cleaner agent.
2. a). What are the constraints Satisfaction Problems? Solve the following Crypt-arithmetic Problem. 7

	C	R	O	S	S
+	R	O	A	D	S
D	A	N	G	E	R

   7
   b). Explain with an Example how A* search can find the optimal solution when there is more than one solution? 8
3. a). What are the problems of depth first search and breadth first search? How are these problems resolved? Explain with a suitable example? 7
   b). Consider the following facts: 8
   i) All creatures that live in forests are beautiful, are magical
   ii). Unicorns live in forests.
   iii). The creature named twilight is unicorn.
   iv). Twilight is beautiful.
   a) Convert the facts in predicate logic.
   b). use resolution to prove "twilight is magical".
   c). Write those facts in prolog statements.
4. a). How does Min-conflicts algorithm work? Explain with the example of queen problem. 7
   b). What is machine learning? Discuss the difference between regression, classification and clustering in machine learning. 8
5. a). Explain the k-means clustering algorithm with step-by-step demonstration on a small dataset, showing how the result. 7
   b). Describe the process of Fuzzification and Defuzzification with an example? 8
6. a). What is Artificial Neural Network? Explain the perceptron training algorithm. Highlight the limitation of the perceptron training algorithm and mention a scenario where it might fail to perform well. 7
   b). Provide examples of ambiguity at the phonetic, syntactic, semantic, and pragmatic levels in Natural Language Processing. 8
7. Write short notes on [Any Two] 10
   a. fuzzy logic
   b. Semantic nets
   c. Learning by analogy

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